EmotionSense — 4-Week Project Plan

Team Structure: 3 Members (Frontend Engineer, Backend Engineer, Machine Learning / Computer Vision Engineer) | **Schedule:** 4-Week Sprint

Executive Summary: EmotionSense is an AI-powered facial emotion intelligence and analytics platform. This engineering plan outlines weekly deliverables across three dedicated engineering tracks to deliver a production-ready system within a 4-week development cycle.

Week 1: System Architecture, Data Schema & ML Pipeline Baseline

Sprint Goal: *Establish foundational architecture, database schema, model inference baseline, and UI layout.*

[Frontend Track] Frontend Engineer

- Initialized React + Vite workspace with Material UI (MUI) and custom design tokens (Violet/Indigo #6366f1 palette, dark/light theme switching).
- Implemented responsive application shell (Sidebar, Navbar, and client-side routing).
- Built Authentication UI: Login, Registration, and Input Validation views with toast alerts.

[Backend Track] Backend Engineer

- Configured FastAPI server with database connection pool (LibSQL/SQLite).
- Designed core relational schema (users, uploaded_files, detection_results, emotion_statistics, model_feedback).
- Implemented secure authentication with BCrypt password hashing and JWT token generation (/register, /login, /profile).

[ML / CV Track] Machine Learning Engineer

- Evaluated and selected deep learning emotion recognition backbones (DeepFace / CNN for 7 basic emotion classes).
- Built preprocessing pipeline: OpenCV face detection, normalization, grayscale/RGB alignment, and frame-rate throttling.
- Validated inference pipeline on static test image datasets to measure baseline latency and accuracy.

> **Key Milestone Deliverable: Functional authentication system, UI dashboard skeleton, and working standalone Python script executing facial emotion classification on test images.**

Week 2: Core Processing, Uploads & Real-Time Emotion Analytics

Sprint Goal: *Connect frontend and backend to support live webcam feeds, file uploads, and chart visualizations.*

[Frontend Track] Frontend Engineer

- Developed media upload interfaces with drag-and-drop validation for images and video formats.
- Implemented WebRTC-based Live Camera capture interface with real-time video stream overlay.
- Integrated Recharts visualization components (Emotion distribution radar/bar charts, confidence gauges).
- Created AuthContext with Axios interceptors for authenticated API requests.

[Backend Track] Backend Engineer

- Implemented multipart file upload endpoints (/upload) supporting images and video clips.
- Developed video frame parsing and background task orchestration for batch inference.
- Created session aggregation endpoints: /history (with search and pagination) and /analysis/{file_id}.

[ML / CV Track] Machine Learning Engineer

- Engineered emotion metrics algorithms: Stability Score Formula (measuring emotional variance across frames), Dominant Emotion Derivation, and Stress & Composure Indices.
- Optimized video frame sampling rate (2-5 FPS) to maintain real-time performance without GPU bottlenecks.

> **Key Milestone Deliverable: End-to-end working system where a user uploads a video or runs a live session and receives comprehensive emotion telemetry and visual breakdown graphs.**

Week 3: Advanced Features, Comparative Intelligence & Admin Portal

Sprint Goal: *Implement side-by-side session comparisons, ground-truth correction feedback, and administrative controls.*

[Frontend Track] Frontend Engineer

- Built Side-by-Side Session Comparison Page (CompareView): Dual session dropdown pickers with swap capability, Executive Delta KPI Scorecards, and GroupedEmotionBarChart.
- Implemented History table multi-selection and floating action bar for instant comparison triggering.
- Built Administrative Portal UI (user tables, audit activities, data purge modals).

[Backend Track] Backend Engineer

- Implemented /api/v1/analysis/compare endpoint calculating analytical deltas across any two session IDs.
- Developed Administrative API suite: /admin/stats, /admin/users, /admin/activity, /admin/purge-activity, /admin/reset-platform.
- Implemented role-based access control (RBAC) middleware verifying admin privileges.

[ML / CV Track] Machine Learning Engineer

- Developed Ground-Truth Feedback System (/feedback): Enabled users to flag misclassified frames with corrected emotions.
- Designed error tracking and confusion matrix aggregation (/admin/feedback) to pinpoint model confusion pairs (e.g., Sad vs. Neutral).
- Tuned confidence thresholds to suppress false positives on edge angles and lighting variations.

Key Milestone Deliverable: Full side-by-side comparison capability, human-in-the-loop ML correction collection, and admin management dashboard.

Week 4: Security Hardening, System Optimization & Final Polish

Sprint Goal: *Token migration, bug fixes, end-to-end integration testing, and production deployment preparation.*

[Frontend Track] Frontend Engineer

- Migrated authentication tokens and admin payloads from localStorage to sessionStorage across axios.js, AuthContext.jsx, and AdminContext.jsx for session isolation.
- Polished UI micro-interactions, responsive mobile views, empty states, and standardized GMT+6 datetime formatting.
- Conducted frontend production build optimization (vite build chunk splitting and asset optimization).

[Backend Track] Backend Engineer

- Resolved FastAPI route ordering conflicts (ensuring /analysis/compare takes precedence over parameterized paths).
- Added database indexing on foreign keys (file_id, user_id, created_at) for sub-second query performance.
- Validated cascading deletion workflows across user files, detection logs, and statistical records.

[ML / CV Track] Machine Learning Engineer

- Conducted comprehensive benchmark validation across test datasets; documented model accuracy, latency, and failure cases.
- Optimized memory footprint during batch video inference and prevented memory leaks in OpenCV video capture loops.
- Packaged model artifacts and inference dependencies for reproducible environments.

Key Milestone Deliverable: Production-ready full-stack application with automated testing, session storage security, side-by-side comparison analytics, and comprehensive project documentation.

Executive Sprint Summary Matrix

Week	Frontend Focus	Backend Focus	ML / CV Focus
Week 1	Vite+React, MUI, Theme tokens, Auth UI	FastAPI, SQLite schema, JWT Auth endpoints	Face detection, OpenCV pipeline, 7-emotion baseline
Week 2	Media uploads, Live WebRTC feed, Recharts	Multipart uploads, Video parsing, Analysis API	Stability score, Stress index, Real-time FPS tuning
Week 3	Side-by-Side Compare UI, Delta KPIs, Admin Portal	/analysis/compare engine, RBAC, Admin analytics	User feedback loop (/feedback), Error matrix analysis
Week 4	sessionStorage security, UI polish, Vite build	Route fix, DB indexing, Cascading deletion purge	Latency benchmarking, Memory optimization, Validation